SHINVA 新华医疗

Automatic Washer-disinfector
Operation Manual

Rapid-M-320

山东新华医疗器械股份有限公司
SHINVA MEDICAL INSTRUMENT CO.,LTD.

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1. Introduction

This manual includes important information regards to routine operation of Rapid-M-320 washer-disinfector. It is necessary for any operator that reads and understands the manual carefully. Since this manual is very important for operators' health and safety, please store it in reachable place. This manual only applies to Rapid-M-320 washer-disinfector.

Product unpacking and connection instruction and drawing are attached.

To ensure the operators are fully trained in safety using aspects, we suggest:

All operators should be trained in the aspect of product operation and maintenance;

For those operators who may contact chemicals and steam, chemicals leaking test knowledge should be learned;

Regular training and examination should be given to operators and maintenance staff to evaluate their knowledge in product operation;

A complete product preventative maintenance plan and safety standards are very important, please strictly comply with all the requirements that included in this manual to bring the maximum performance into full play.

The following symbols in this manual are used to get highly attention and caution, please read carefully.

Highly attention should be paid.

The cases possibly cause equipment damage.

The cases possibly endanger personal safety.

Warning: danger of scald.

Do not open any pipeline during the operation. Hot water may extrude out.

Do not inspect the equipment until the pipeline system cools down.

Warning: danger of tumble.

To avoid tumble, please keep the floor dry.

Warning: danger of personal safety or equipment damage.

To ensure the equipment operates reliably, both regular and preventative maintenance are necessary.

Equipment repair and adjustment must be done by specializedperson. Any unqualified maintenance or spare parts replacement may cause personal injury.

Warning: danger of electric shock or scald.

Please cut off all the water, power and steam supply before maintenance and all the supply must be cut off properly.

Fasteners and gaskets are used for connection, please make sure all the fasteners and gaskets are fixed back after maintenance.

Warning: danger of chemical burn or eye injury.

To avoid eyes or skin injury please wear gloves and eye shield when use descaling product. If the chemical liquid splashes on skin please promptly wash by clean water for 15 minutes. If accidently swallowing happens please drink alkaline substance with plenty of clean water or see doctor.

The cleaning agent is corrosive to skin, please keep it away from eyes or skin and do not swallow it.

Please read and comply with the preventative measures and instructions labeled on the cleaning detergent bottle when deal with detergent pump or other cleaning detergent containers.

Please ware gloves, face mask and protective clothing when deal with cleaning detergent or repair detergent pump and pipeline.

Caution

Please choose low chloride content cleaning detergent to avoid injury on stainless steel.

Please wipe the equipment by soft cloth to avoid scratch on stainless steel.

If the equipment is installed beside other equipment please identify matching valve and power switch to disoperation.

Shout down the switch in any emergent situation to terminate the equipment.

Important notification:

According to hospital standards, after cleaning and disinfection please double check the instruments if there is any residual dirt. It is proved that the entire visible residue can be exterminated under proper operation.

1 Please double check each instrument if there is any residual dirt.

2 Please rehandle those instruments that has visible residue.

2. Operating principles

Rapid-M-320 automatic washer-disinfector is high efficient manual-door washing equipment that researched and developed according to hospital actual situation. It is widely used for disinfection of surgical instruments, wares, medical trays and plates, anesthesia instruments and breath tubes in hospital central sterile supply department or operating room. The greatest advantage of the equipment is fast washing speed that could shorten 1/3 operation time.

Softened and pure water, which used as disinfection medium, circulate in chamber washing pipeline through the circulating pump. The washing detergent is automatically injected while the rotating arm sprays the water on the instruments to wash them. Also, the water and air could be heated automatically for sterilization and drying program.

3. Operating conditions

a) Power source: 380VAC, 50Hz

b) Environment temperature: 5~40℃

c) Relative humidity: ≤90% (in 20℃)

d) Steam source: 0.3~0.5mpa flow rate: 100kg/h

e) Water source: 0.2~0.5mpa flow rate: 100L/h

f) Softened water source: 4-7ºd（1 ºd=0.178mmol/L）

g) Pure water source: ≤15μm/cm (in 20℃)

4. Operation

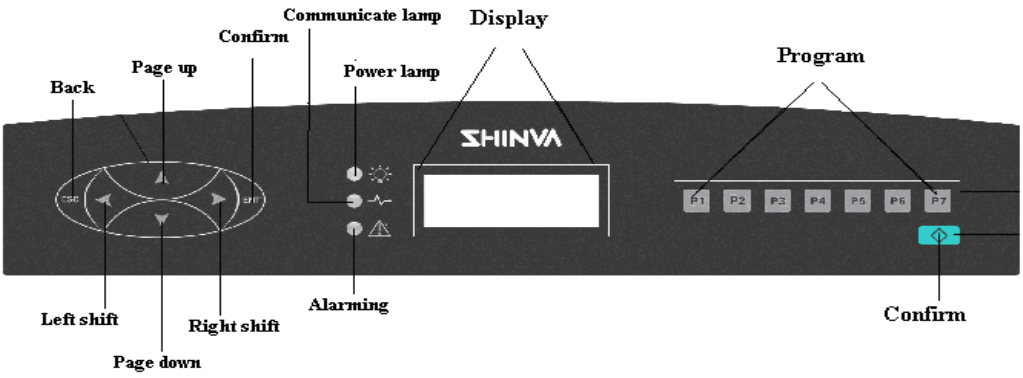
4.1. Door operation

1 Please turn the handle to open the manual door.

2 The door cannot be opened during operation.

3 As a partition wall between polluted and clean area,the front and back door cannot be opened at same time

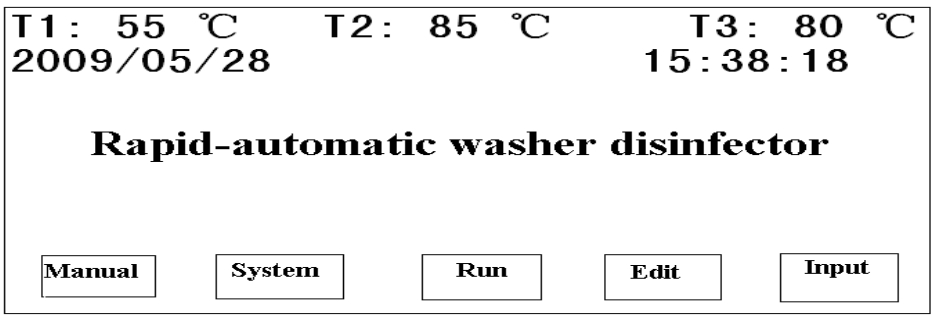
4.2. Program running



p1

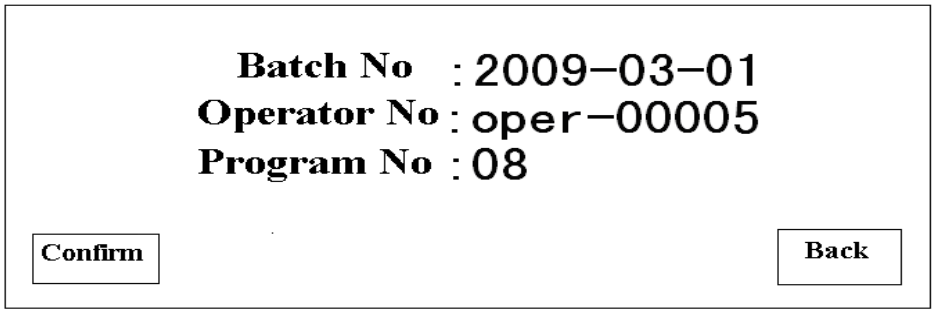
4.2.1. Program running interface

Turn on the power you will see the screen displays as **P2**



P2

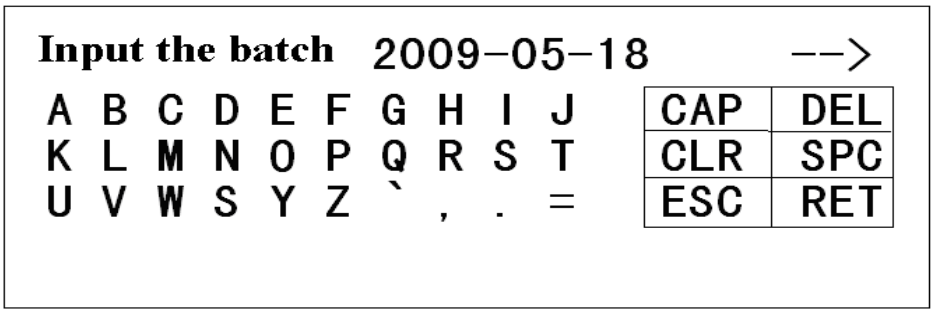
Move the cursor to 'Run' and click 'Confirm', the screen displays as P3



P3

4.2.2. Batch number input

Move the cursor to the batch number input field by pushing the cursor keys and push'Confirm' key to enter in P4. After inputting the batch number please push 'RET' key to return P3.



P4

4.2.3. Operator number input

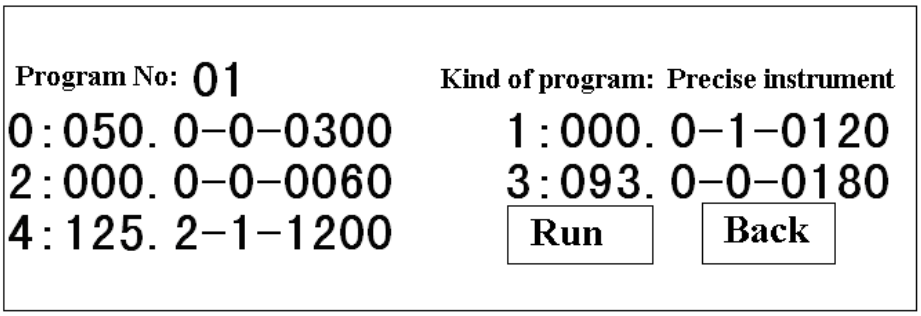
Move the cursor to the operator number input field by pushing the cursor keys and push 'Confirm' key to enter in P4. After inputting the operator number please push 'RET' key to return P3.

4.2.4. Program number input

Move the cursor to the program number input field by pushing the cursor keys and push ‘Confirm’ key to enter in P4. After inputting the program number please push ‘RET’ key to return P3.

4.2.5. Program running

After inputting relevant information please push ‘Confirm’ key to enter in P5.



P5

P5 will display the selected program number, type and parameters of each stage. Now push ‘Run’ key then push ‘Confirm’ key to initiate program operating condition test. The program will be initiated (and enter in P6 automatically) only if all the conditions are met, otherwise the alarm will prompt the operator.

Program operating conditions:

1 Both front and back door are closed.

2 Do not leave the selected parameter field blank.

3 All temperature sensors are in normal status.

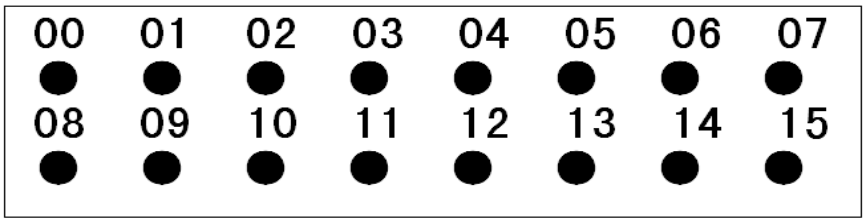
System initialization:

1 To initiate the first 10 programs parameters, please type ‘add’ in P3 as operator number. The system will prompt ‘SUCCESS’ when the initialization finished successfully.

2 For factory setting, please type ‘clearram’ in P3 as operator number. The system will prompt ‘SUCCESS’ when the setting finished successfully.

4.3. Input monitoring

Push the ‘Input’ key to enter in P6.

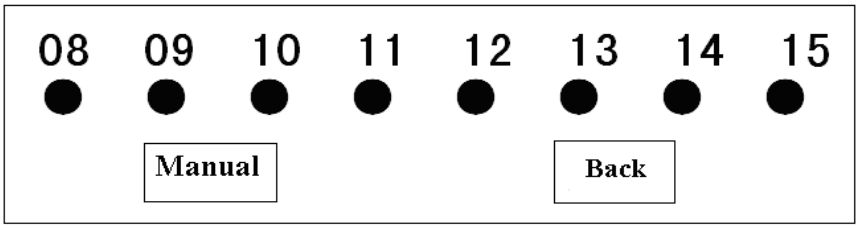


P6

P6 shows the status of all monitoring components. If there is an input signal the corresponding dot will turn to black. To return to the main interface push 'Back' key.

4.4. Manual operation

Select 'manual' in P2 and push 'Confirm' key to enter in P7.



P7

Manual/auto shift

Select 'Manual' or 'Auto' by pushing the cursor keys and push 'Confirm' key. Select Manual' then select the output to be operated and push 'Confirm' key. The corresponding dot will turn to black. Push the 'Confirm' key again to exit.

The corresponding manual outputs are as follows:

00 front door lock 06 hot water valve open 11 buzzer 16 front door light

01 back door lock 07 dewatering valve 12 air heating tube 17 operation light

02 softened water valve 08 cleanser pump 13 circulation pump 18 termination light

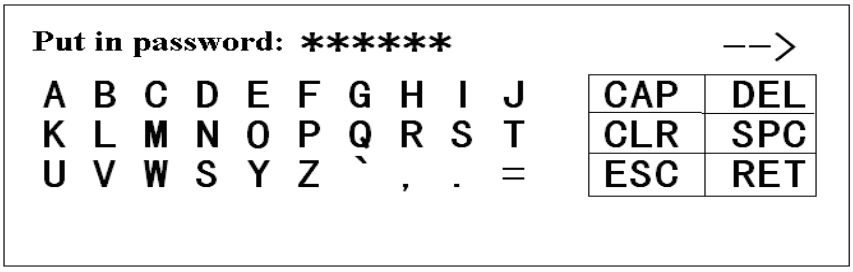
03 pure water valve 09 oil supply 14 air blower 19 hot water valve close

05 water tank injection valve 10 disinfectant pump 15 chamber heating tube

20 water tank heating valve

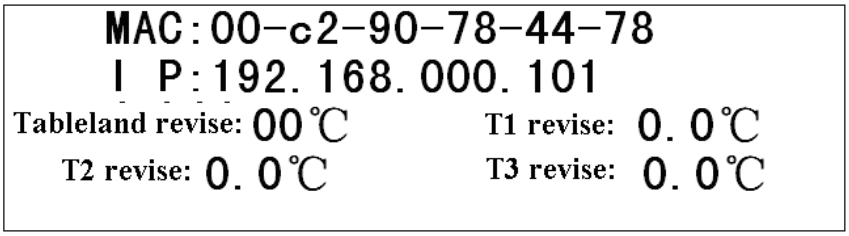
4.5. System setting

Select 'System' in P2 and push 'Confirm' key to enter P8.



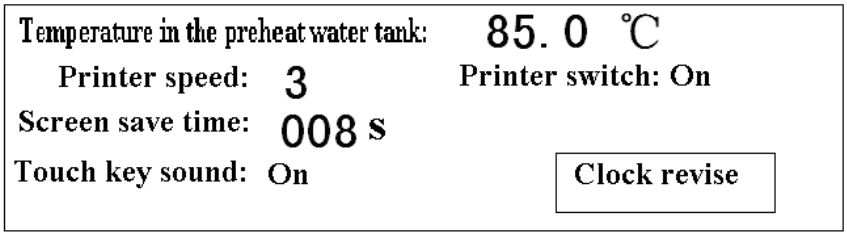
P8

Enter the correct password (initial password is 78) and push 'RET' key to enter in P9.



P9

Push 'page down' to enter in P10.



P10

4.6. Parameter instruction

a. T1 adjustment: to adjust chamber temperature platinum thermistor.

Formula: chamber temperature T1 = chamber detecting temperature + T1 adjustment

b. T2 adjustment: to adjust air temperature platinum thermistor.

Formula: air temperature T2 = air detecting temperature + T2 adjustment

c. T3 adjustment: to adjust preheating water tank temperature platinum thermistor.

Formula: preheating water tank temperature T3 = preheating water tank detecting temperature + T3 adjustment

d. The preheating water tank temperature is the final treated temperature of the preheating water tank. The default value of that is 85℃ and the setting range is 60-86℃.

e. The printer status can be changed to turn on or off the printer.

f. The ‘printer speed’ has 3 options as follows:

speed option 1: 1cm/5mins

speed option 2: 1cm/10mins

speed option 3: 1cm/15mins

g. The ‘plateau adjustment is a for plateau effect to deal with the case that the water temperature in plateau area cannot reach certain temperature. The default value of that is 0℃.

4.7. Parameter setting

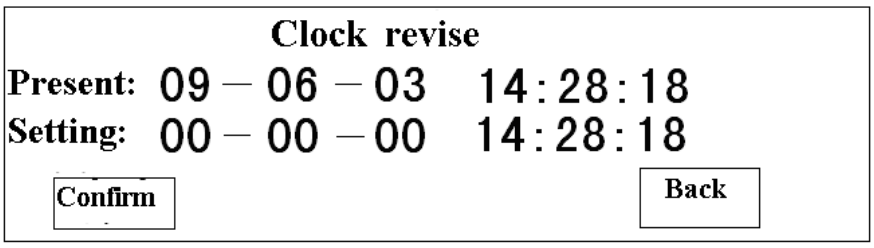
a. Move the cursor to parameter that to be set then push ‘Confirm’ key to edit it. Push arrow key to increase or decrease the value. To finish parameter setting please push ‘Confirm’ key.

b. P9 and P10 can be shifted by pushing the arrow key.

c. Push ‘Back’ key to go back to the main interface P2.

4.8. Clock revise

Move the cursor to ‘clock revise’ key in P10 and push ‘Confirm’ key to enter in P11.



P11

Edit the time by moving the cursor. Push ‘Confirm’ key to finish the time setting.

Push ‘Confirm’ key to active the setting.

Push ‘Back’ key to go back to the main interface P2.

4.9. 4.9 Program parameter instruction

The capacity of the control system is 26 programs. The first 11 programs (program No.

00-10) are set before sale, while the rest can be edited by users.

4.9.1. Program type

Select the corresponding program according to the type of items that to be washed. Please refer to the following instructions:

A—— surgical instrument

B—— precision instrument

C—— ware

D—— plastic item

E—— glass item

F—— anesthesia instrument

G—— others

H—— seriously polluted item

I—— infectant

J—— oil disinfection

K—— disinfection program

L—— drying program

4.9.2. Stage name

The program will display the name of each stage automatically. The stage name can be input manually.

In drying stage, set ‘coldown’ to execute cooling operation which means the air temperature will be decreased below certain value. Only dryer is working in this situation.

4.9.3. Time type

The instructions of 4 time types are as follows:

1—— the timer works as the program starts

2—— the timer works as the water level reaches certain value

3—— the timer works as the temperature reaches certain value

4—— the timer only works as the temperature ≥ certain value

4.9.4. Other parameters

Time

The number is shown in second.

Temperature setting

The number is shown in ℃.

Working medium

There are two working types (‘tank’ and ‘air’) in the program. Click ‘tank’ for water injection or click ‘air’ for drying/air cooling.

Washing liquid 1—peristaltic pump 1 working time. The number is shown in second.

Oil supply liquid—peristaltic pump 2 working time. The number is shown in second.

Washing liquid 2—peristaltic pump 3 working time. The number is shown in second.

Water-in valve—the working valve for this phase.

There are 3 types of water-in valve as follows:

‘CW’ (cold water)—cold water valve

‘WW’ (warm water) —warm water valve

‘CWW’ (cold warm water) —cold/warm water

Drainage—the drainage valve will keep opened for seconds when there is no water level signal. The number is shown in second.

Pre-drainage—the pre-drainage time before each stage. The number is shown in second.

Pump time-delay—the pump will keep getting the water flow in for second. The number is shown in second.

4.9.5. Program setting instruction

Push the ‘Edit’key in P 2 and type in the password ‘78’ to enter into P 12.

Program: 00 Serious contaminant		Step: 00	Step-new
Time: 00 second	Temperature: 123.5℃	Medium: air	
Lotion 1: 0000 seconds	Coating oil: 0000 seconds		
Lotion 2: 0000 seconds	— ← → + ↓		

P12

Program: 00 Serious contaminant		Step: 00	Step-new
Water feed valve: CW	Drainage: 0000 seconds		
Pre-drain: 0000 seconds	Pump retain: 0000 seconds		
Time type: 4	— ← → + ↓		

P13

There are 5 function options in P12: '—', '←', '→','+', '↑'.

'←' and '→' are used for viewing each phase of the program;

'—' is used for deleting current phase;

'+' is used for adding a phase;

Push'↑' to enter into P13 when finish editing.

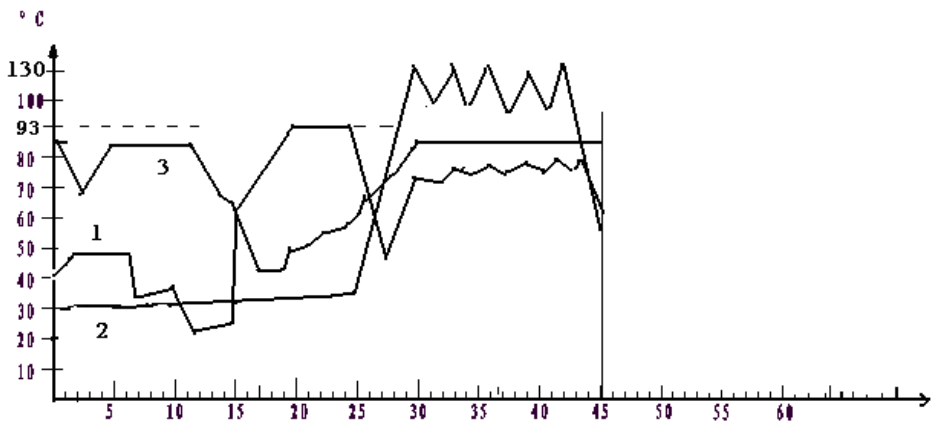
When finish editing parameters in P12 and P13 click '←'/'→' to edit parameters in other phases.

Push 'Back' key to back to P2.

5. 5 Program flow instruction

5.1. 'Instrument' program start and flow control

'Instrument' program working flowchart



5.1.1. Run-up stage

In the run-up stage, the program can be started only if the following matters are achieved:

•The program is in ‘auto’ status.

•The door is in ‘close’ position.

•The door is in ‘seal’ status.

•No water in the chamber.

•The steam pressure is normal.

•The water pressure is normal.

•All temperature transmitters are normal.

If those matters are not achieved, there will be alarm message when you push ‘start’ button. When this stage is finished operator can start the program by pushing ‘start’ button.

5.1.2. Prewashing stage

The prewashing stage defaults to off. To start this stage please push ‘start’ button. Pre-washing is generally used for instruments that are not fully pretreated.

The water inlet valve opens up and the soften water flows into chamber through pipeline system. The water inlet valve will be closed in 5 seconds when the chamber detects water level signal and the circulating pump starts working. The system will enter into next stage when drainage is finished.

5.1.3. Washing stage

This stage called washing or enzyme washing. The water inlet valve opens up and starts injecting water into chamber for 25 seconds. Now the circulating pump starts working. When

the chamber detects water level signal the water inlet valve will close up and the heating tube starts heating the water. When the water temperature goes up to 45℃ the metering pump starts working to supply the cleaning fluid into chamber. The heating tube will stop heating till the water temperature climbs to 45℃. After 300 seconds circulation the drainage valve starts discharge water for 60 seconds and then closes up.

5.1.4. Rinse 1 stage

The water inlet valve opens up and starts injecting water into chamber for 25 seconds. Now the circulating pump starts working. When the chamber detects water level signal the water inlet valve will close up. The water in chamber starts being circulated and the build-in timer starts to time. After 1 minute the circulating pump stops working. Now the drainage valve opens up to discharge water for 60 seconds and then closes up.

5.1.5. Rinse 2 stage

The water inlet valve (F1, F2) open up and starts injecting water into chamber for 25 seconds. Now the circulating pump starts working. When the chamber detects water level signal the water inlet valve will close up. The water in chamber starts being circulated and the build-in timer starts to time. After 1 minute the circulating pump stops working. Now the drainage valve opens up to discharge water for 60 seconds and then closes up.

5.1.6. Disinfection stage

The distilled water inlet valve opens up and starts injecting water into chamber for 25 seconds. Now the circulating pump starts working. When the chamber detects water level signal the water inlet valve will close up. The metering pump injects disinfecting oil into chamber and heating tube starts working. The water in chamber starts being circulated and heated. The build-in timer starts to time when the water temperature climbs to 93℃. The heating tube will be controlled by water temperature from now which means the heating tube stops working when the water temperature exceeds 93℃, while starts working when the temperature comes below 93℃. After 150 seconds both circulating pump and heating tube stop working. The drainage valve opens up to discharge water for 60 seconds.

5.1.7 Drying stage

The draught fan starts working to flow the cold air into air heating tank and air heating tube

starts to work. The heated air comes into chamber to push the wet and hot air out of chamber. This process keeps the chamber temperature climbing up to 90℃. After 15 minutes the air heating tube stops working and the draught fan will stop working in 60 seconds.

6. Door operation instructions

6.1. Basic operation

Back to initial interface and turn the handle to open the door and the door can be a bench when it is opened. The washing rack can be pushed into chamber through the track on the door bench.

After each program there will be system prompt of opening door. Now open the door and pull out the washing rack.

6.2. Front and back door interlocking

This washing product is used in hospital CSSD to separate the clean zone and contaminated zone. It adopts front and back door interlock function which means two doors cannot be opened at same time. This function achieves the isolation between two zones so that avoids secondary contamination.

7. 7 Washing rack and washing accessories operation

7.1. Washing rack instruction

The washer-disinfector cannot come into full play without washing racks. To achieve best washing result, different washing function should be working with certain washing racks.

Items that suitable for washing: general surgical operation instrument, minimally invasive surgical instruments, dressing bowl, kidney plate, beaker, flask, measuring cylinder, test tube, dental instruments, breathing and anesthesia tube, operation shoes, bedpan and feeder, etc.

The items that to be washed should have the following features: 1. Reusable. Any disposable items cannot be washed or autoclaved; 2. Can be wash-disinfected. In general, PVC

material instruments are not suitable for autoclaving, such as breathing and anesthesia tube; plastic instruments cannot be washed by alkaline cleaner; and aluminum instruments cannot be washed by alkaline or acidic cleaner; 3. Items that suitable for washing means those items should be suitable for proper washer-disinfector and washing racks.

Transporting tool: it is used to carry and transport washing rack to different places such as packing table and assorting table.

Loading tool: It is the carrier of washing racks and items that to be wash-disinfected. To achieve best washing result, different washing function should be working with certain loading tools and washing racks.

7.2. Washing rack classification and usage

7.2.1. Four-layer fixed instrument washing rack

This four-layer fixed washing rack is standard configuration of our washing product. It is mainly used for disinfection of surgical operating instrument. This type of washing rack can load two DIN standard trays (480x250x50) on each layer.

Before washing, put the trays that are full of instruments on the rack and push the rack into chamber. Close the door and start operation.

Also, dressing bowls and kidney dishes can be put in between two layers. However, we do not recommend put too many bowls or dishes in there since that will affect washing result.

Four-layer fixed instrument washing rack

7.2.2. Four-layer adjustable instrument washing rack

The use of this rack is the same as fixed rack (see 7.2.1). The distance between two layers can be adjusted and the barrier on the rack can be taken off for washing big size instruments.

7.2.3. Five-layer fixed instrument washing rack

The use of this rack is the same as fixed rack (see 7.2.1). The distance between two layers is relatively small which only can be used for surgical instrument. However, this kind of rack has bigger capacity which can totally load 10 DIN standard trays.

7.2.4. Five-layer adjustable instrument washing rack

The use of this rack is the same as fixed rack (see 7.2.1). The distance between two layers can be adjusted and the barrier on the rack can be taken off for washing big size instruments.

Five-layer washing rack

7.2.5. Anesthesia/breath tube washing rack

This kind of rack is mainly used for washing anesthesia tube, breath tube and accessories. Before using this washing rack please read the using direction that provided by anesthesia/breath tube manufacturer to make sure the tube and accessories can be washed in this rack. Firstly, please make sure the material is heat resisting. Generally, PVC material is not suitable for high temperature disinfection. Secondly, please refer to the using direction for dismantling the tube. Proper dismantling will improve the disinfecting effect.

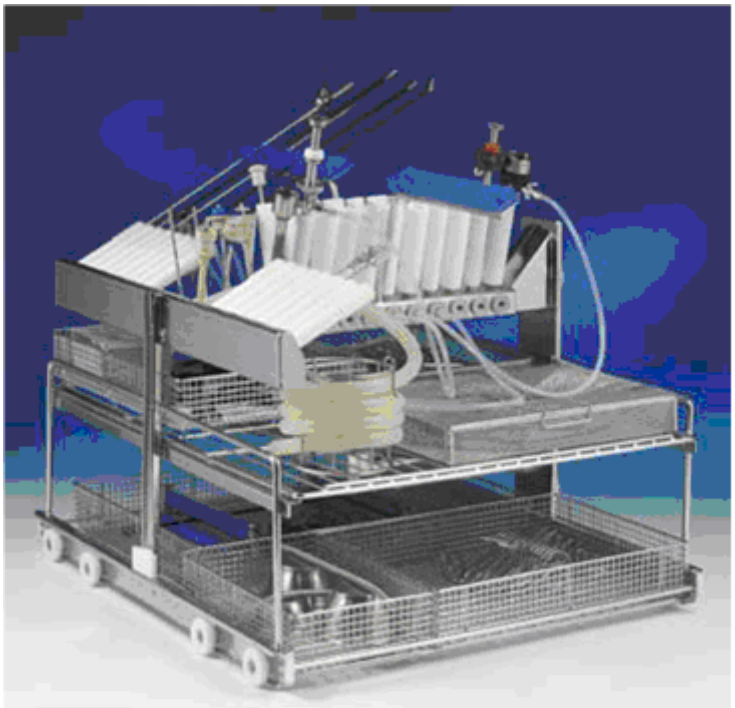
There are three rows of bent spray tubes fixed on the rack for washing different length of anesthesia/breath tube. Put the anesthesia/breath tube on the bent spray tube of washing rack. We recommend put the longer anesthesia/breath tubes on the outer side of spray tube, while put the shorter ones on the inner side. The straight spray tubes that fixed in the middle of the rack are used for washing breathing bags, humidification bottles or other medical containers. The position of t-frame support on the bottom of the straight spray tubes can be adjusted. There are two rows of tiny spray tubes for washing small-sized instruments such as t-branch pipe.

Anesthesia/breath tube washing rack

7.2.6. Minimally invasive surgery instrument washing rack

This kind of rack is mainly used for washing rigid endoscope, puncture needle and aspirator, etc. Please read the using direction for dismantling the tube. Proper dismantling will improve the disinfecting effect.

This washing rack can load three sets of endoscopes. It adopts independent fix support which improves the flexibility of washing and disinfection.



Minimally invasive surgery instrument washing rack

7.2.7. Dental handpiece washing rack

There are many silicon bases on the washing rack for holding dental handpiece, and each base is connected with a water inlet tube. The circulating water will flow through high pressure circulating pump and spray to the inside of dental handpiece to wash it with specialized cleaning fluid. This washing rack also equipped with choke plugs which are used for those bases without holding dental handpiece.

This rack has three layers: the first layer, which has 36 silicon bases, is applicable forφ19 dental handpiece; the second layer, which has 36 silicon bases, is used forφ15 dental handpiece; the third layer has two trays for washing other instruments at the same time.

Before washing, put the dental handpieces on the matching silica gel bases and insert the choke plugs into the empty bases if any. Now push the washing rack into chamber and start the program.

7.2.8. Glass ware washing rack

This kind of washing rack is mainly used for washing glass-made instruments such as beaker, wild-mouth bottle, boiling tube and beaker flask, etc. Before washing, place the wares on the spray tubes in upside-down way. The water will be sprayed out from the spray tubes and spray arms to wash the inner and outer surface of glass wares.

Please offer us the accurate rack size when you order the customized washing rack.

Glass ware washing rack

7.2.9. Humidification bottle washing rack

The working principle and use of this kind of washing rack is the same as glass ware washing rack (see 7.2.8). The only difference is that this rack is used for washing humidification bottle. It has two layers that could load 160 humidification bottles. The standard rack is designed for medium-sized humidification bottles.

7.2.10. Medical bowl/dish washing rack

The disinfection of dressing bowl and kidney dish is a main use of our EASY series and Rapid series disinfector. On the basis of this purpose we developed this three-layer washing rack.

The first layer can load two DIN standard trays or three (480×250×50), while the second and third layer can be used for dressing bowls and kidney dishes. To increase the rack capacity and ensure the washing effect, please put the bowls and dishes properly.

The following sizes of dressing bowl are suitable for this rack:

φ160×60; φ142×55; φ125×45

Please offer us the accurate size if you order customized washing rack for washing other size of bowls.

The following sizes of kidney dishes are suitable for this rack:

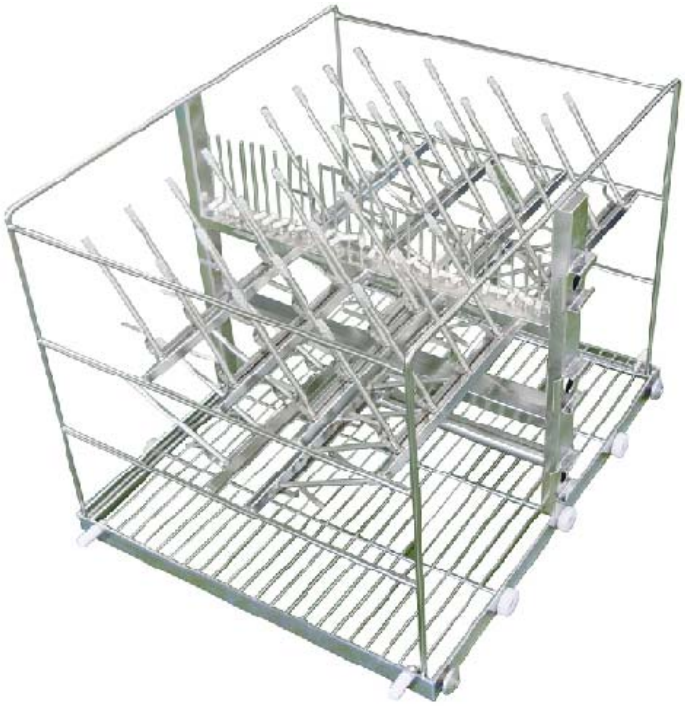
204×130; 190×116; 125×45

Please offer us the accurate size if you order customized washing rack for washing other size of dishes.

Medical bowl/dish washing rack

7.2.11. Vatting washing rack

The use of this kind of washing rack is the same as glass ware washing rack (see 7.2.8).



Vatting washing rack

7.2.12. Customized made washing rack

Please offer us the accurate rack size when you order the customized washing rack.

7.3. Other washing accessories operation

7.3.1. Transport trolley

It is used for loading and transporting washing racks. The auto positioning equipment between washing rack and transport trolley makes it convenient to push the rack into chamber. The trolley also equipped with four universal wheels and two wheel brakes.

7.3.2. Washing rack storage table

It is used for long-time storage of washing rack or other articles.

Washing rack storage table

7.3.3. Instrument tray

At present, there are two types of standard instrument trays in the field of washing-disinfector: DIN (Deutsche Industrie Norm) standard instrument trays and SPRI standard instrument trays. DIN standard instrument tray includes four sizes: 480×250×50; 240×250×50; 480×250×70;

240×250×70.

The size of our standard trays that equipped with the Rapid-M-320 is 480×250×50 which is the most popular size. We also offer optional size of 240×250×50.

The SPRI standard instrument trays includes three sizes: 450×340×70; 340×250×70; 250×170×70.

8. User authorization management

The user authorization management is achieved through button security level. Once the button is protected, the corresponding link cannot be accessed without correct ID number and password.

Administrator

Administrator has authority of operation and viewing. It has following functions:

1) User management
2) Parameter setting
3) Program editing
4) Initialize system

Maintenance staff

The maintenance staff has authority of routine maintenance. It has following functions:

1) Deviation correction
2) Time correction
3) Manual operation

Operator

The operator has authority of distributing operator authority and operating those button that has no security control. It has following functions:

1) Door operation. i.e. open/close door
2) Program running

3) Check alarm message
4) View input signal
5) Language select

9. Alarm message verification

No.	Information	Instruction	Solution
1	Front door is not closed, program cannot be started.	Front door is not fully closed.	1 Close the front door. 2 Check the travel switch.
2	Back door is not closed, program cannot be started.	Back door is not fully closed.	1 Close the back door. 2 Check the travel switch.
3	Program is not edit-ed.	Program is not edited.	1 Select a program which is ed-ited and start it. 2 Initialize the program.
4	Water injection slow.	Water cannot reach certain level within certain time.	1 Check whether the water pressure is normal (0.2-05Mpa). 2 Check whether the water valve is open. 3 Check whether the water valve filter is blocked. 4 Check whether the water tank filter is blocked.
5	Chamber water level low.	Water level switch cannot detect water level.	1 Check whether the chamber water level is normal. 2 Check whether there is leakage in the chamber. 3 Check whether the water injec-

			tion is normal.
6	Blast pressure low.	The pressure of air outlet is low.	1 Check whether the fan is working. 2 Check whether there is fault on drying pipeline. 3 Check whether the pressure-leading tube is properly connected. 4 Check the set point of switch.
7	Water in drying system.	Hot air tube has detected water level signal.	1 Check whether the drying one-way valve is working. 2 Check whether there is any fault on floater of hot air tube.
8	Heating up slow.	The temperature cannot reach certain level within certain time.	1 Check whether the heat valve is open. 2 Check whether the filter is blocked. 3 Check whether the steam pressure is too low. 4 Check whether there is any fault on heating tube.
9	Drying blower protection.	The fan cannot work normally.	1 Check whether the blower works properly. 2 Check whether there is any default phase. 3 Check whether the blower is overloaded.
10	Probe failure.	System cannot detect temperature signal.	1 Check the probe. 2 Check the circuit.

			3 Check the module.
11	Water is not ready.	The temperature of pre-heating water tank does not reach certain value.	1 Check the water supply and water pressure. 2 Check the steam supply (only for the product which uses steam). 3 Check whether the heating valve is open under heating status. 4 Check whether the heating tube is working.
12	The water in chamber is not discharged totally.	There is residual water in chamber.	1 Check the drainage valve. 2 Check whether the drainage pipeline is blocked. 3 Check the water level switch is working normally. 4 Check whether the there is water in pressure-leading tube.
13	Please inject cleaning fluid.	Cleaning fluid signal cannot be detected.	1 Inject cleaning fluid. 2 Check whether the float bowl of water level detecting switch is properly installed.
14	Please inject oil agent.	Oil agent signal cannot be detected.	1 Please inject oil agent. 2 Check whether the float bowl of water level detecting switch is properly installed.

10. User operation standards

10.1. Preparation work

•Make sure there is no water in chamber.

•Turn on the water supply and make sure the water pressure is between 0.2~0.5 Mpa.

•Turn on the steam supply and make sure the steam pressure is between 0.3~0.5Mpa.

•Turn on the master control switch.

•Please inject cleaning agent if system prompts lack of cleaning agent.

•Turn on the supply of water, power, steam and compressed gas half hour in advance to get hot water ready for operation.

10.2. Program running

•Put the full load washing rack into chamber and start program.

•The washing rack should be put properly in chamber to make sure the spray arms are not blocked.

•After washing please pull the washing rack out from the back door of chamber.

10.3. Termination

•The system will be terminated after a program cycle. Reset the program to start next cycle.

10.4. After-work matters

•Turn off the supply of power, water, steam and compressed gas.

•Clean up the chamber every day after work. Maintain the chamber weekly and monthly.

•Examine and repair the product weekly to make sure there is no any leakage and loose switch. Check up the rotation of spray arms.

10.5. Attentions

•In manual operation, the circulating pump cannot work without water.

•In auto/manual operation, the water heating tube starts working only if the system detects water level signal.

•In auto/manual drying operation, the air heating tube starts working only if the fan is working. After turning off the heating tube, the fan should be turned off after a cooling period.